Signal iOS 8.37 (build 1993)

Source-to-binary verification of the App Store build

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Result. The compiled code (`__TEXT,__text`) of all 21 binaries in the App Store app is **byte-identical** to a local build of tag `8.37.0.1993` with Xcode 27.1 (27A9275). That covers Signal's five own binaries and all 16 bundled frameworks. Every other difference is one already explained for earlier versions: the signature-size field in every binary; the UUID and the linker's GOT-slot order in `Signal` and `SignalServiceKit` , with every stub resolving to the same target; build metadata in plists; and rasterised images (the app icons and two asset renditions), which differ slightly from identical vector sources.

Its four prebuilt native libraries (libsignal, RingRTC, WebRTC, libmobilecoin) were also accounted for: libsignal `v0.106.1` and libmobilecoin `6.0.2` are unchanged from 8.36 and their rebuilds carry over, and RingRTC `v2.74.0` with WebRTC `8037b` were rebuilt from source and match, so 8.37 is **fully reproducible from public source**. That covers not just the Signal-iOS repository but every library it bundles.

1 Method

This is the same method as for 8.35.1, which describes it in full.

1. The source is cloned at the Xcode Cloud path `/Volumes/workspace/repository` .
2. The third-party frameworks and prebuilt binaries are Swift Package Manager packages pinned with exact versions and resolved by `xcodebuild` (8.37 dropped the CocoaPods `Podfile` / `Makefile`).
3. An unsigned `xcodebuild archive` runs with the scheme's App Store Release configuration, using the exact Xcode build recorded in the shipped `Info.plist` .
4. `verify/compare_build.sh` and `verify/explain_residuals.sh` compare the result with the decrypted App Store IPA.

Input	Value
App Store app	Signal 8.37 (1993), downloaded from the App Store (version ID `894994822`) and decrypted with unfairplay. The encrypted IPA passes all 27,161 of its code-signature page hashes. Decryption changed nothing outside the FairPlay-encrypted ranges apart from the `cryptid` flag.
Source	Tag `8.37.0.1993` , commit `1c925a184` , the commit recorded in the shipped `Info.plist` . libsignal `0.106.1` , RingRTC `2.74.0` and MobileCoin-Swift `6.0.3-signal.1` at the versions the tag pins.
Toolchain	Xcode 27.1 (27A9275), matching the shipped `DTXcodeBuild` .

2 Results

Check	Result
`__TEXT,__text` , all 21 binaries	Byte-identical (69.0 MB, hashed with Apple's `segedit` and `shasum`)
Functions, Signal's 5 binaries	153,121 of 153,121 match after address normalisation
Strings, symbols, linked libraries	Identical

Check	Result
Header bytes	`__LINKEDIT` size (signature) in all 21 binaries; `LC_UUID` in `Signal` and `SignalServiceKit`
GOT slot order	`Signal` (1,693 + 1 stub bytes) and `SignalServiceKit` (998 + 1 stub bytes). All stubs resolve to the same target.
Plists	38 differ, only in `BuildDetails` time, `BuildMachineOSBuild`, `CFBundleVersion` and `DTAppStoreToolsBuild`
Asset catalogs	All renditions identical except the 72 rasterised app-icon bitmaps (renamed randomly by `actool`) and one rasterised rendition each in the main app (`official-wallpaper`) and the share extension (`clock-diver`), whose vector sources are identical
Loose icons	107 of 14,400 and 176 of 23,104 pixels differ, by at most 1/255 in one channel
Prebuilt native libraries	libsignal `v0.106.1` and libmobilecoin `6.0.2` unchanged from 8.36, whose rebuilds carry over; RingRTC `v2.74.0` rebuilt and byte-identical; WebRTC `8037b` rebuilt, `__TEXT,__text` byte-identical and the whole binary differing only in `LC_UUID` and one linker marker. See below.

3 Prebuilt native libraries

libsignal `v0.106.1` and libmobilecoin `6.0.2` are pinned to exactly the binaries verified for 8.36, so no rebuild is needed here and the earlier results carry over (8.36 deps and 8.35.1 deps).

RingRTC `v2.74.0` and WebRTC `8037b` are new pins for 8.37, so they were rebuilt from source and compared with the pinned prebuilds:

- RingRTC `v2.74.0` was built with `verify/deps/build_ringrtc.sh` (Rust 1.97.1, Xcode 26.2 at the runner's paths) and its `libringrtc.a` matches the pinned `ringrtc-ios-build-v2.74.0.xcframework.zip` byte for byte.
- WebRTC `8037b` was built with `verify/deps/build_webrtc.sh` (Xcode 26.1.1) and its arm64 slice matches the bundled `WebRTC.framework` apart from two things: the 16-byte `LC_UUID`, and one `radr://5614542` linker-synthesised marker symbol (with 96 bytes of associated string-table data). The linker emits that marker on Signal's macOS 15 CI host but not on the macOS 26 host used here — a build-host linker artifact, not source code. `__TEXT,__text` is byte-identical.

With the app's own code also byte-identical, every compiled byte in 8.37 traces back to public source: the equivalent of a source-reproducible build. Scripts: `verify/deps/`; outputs: `out/deps/`.

4 Limitations

As for earlier versions:

- The public source itself is not reviewed.
- The toolchains are trusted: Apple's Xcode compilers, the Rust toolchains from rustup and Chromium's prebuilt clang. A compromised compiler would reproduce its own output.
- The causes of the UUID and icon-rasterisation differences are inferred, not proven.

5 Reproducing

```
git clone https://github.com/signalapp/Signal-iOS.git /Volumes/workspace/repository
cd /Volumes/workspace/repository && git checkout 8.37.0.1993
# xcodebuild archive with Xcode 27.1 (27A9275), then:
verify/compare_build.sh signal-ios-8.37.0-1993.decrypted.ipa <built Signal.app>
```

```
versions/8.37.0.1993/out/ipa_vs_build
verify/explain_residuals.sh signal-ios-8.37.0-1993.decrypted.ipa <built Signal.app>
versions/8.37.0.1993/out/residuals
```

The decrypted IPA is attached to the `signal-8.37.0.1993` release, and all outputs are in `versions/8.37.0.1993/out/`.

6 Changelog

Date	Change
12 November 2026	First published version.